Intro

This lab goes over a simple case structure implementation, merging signals, and a simple “power absorbed” calculator.

The application is a rectifier operating on V_S to output V_{out} and it is expected to display both to the use on the same waveform chart.

To calculate power absorbed by R1 it is simply

$$P = \frac{{V_{out}}^2}{R_1}$$

Where V_{out} is the rectified form of V_S based on a conditional:

$$V_{out}(t) = \begin{cases} V_S\,(t) & V_S\,(t) \geq 0 \\ -V_S\,(t) & V_S\,(t) < 0 \end{cases}$$

And R_1 is some resistor value

And

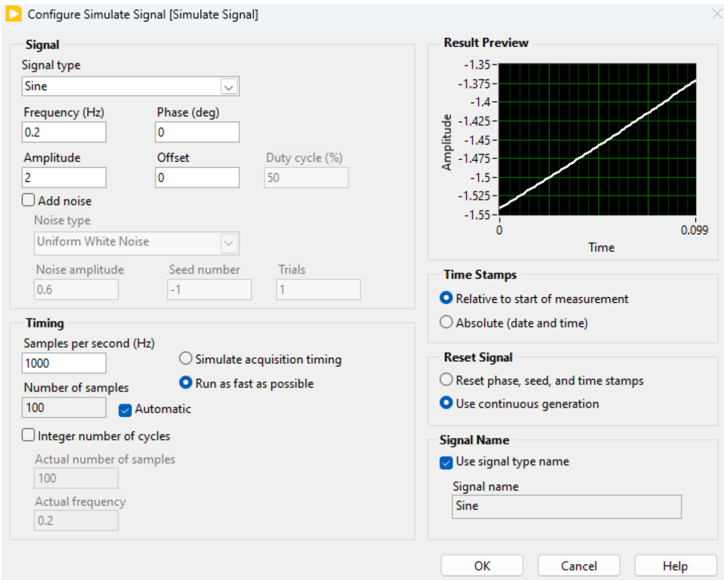
$V_S(t) = 2\sin(2\pi(0.2)t)$

A case structure acts much like an if-else statement in implementation.

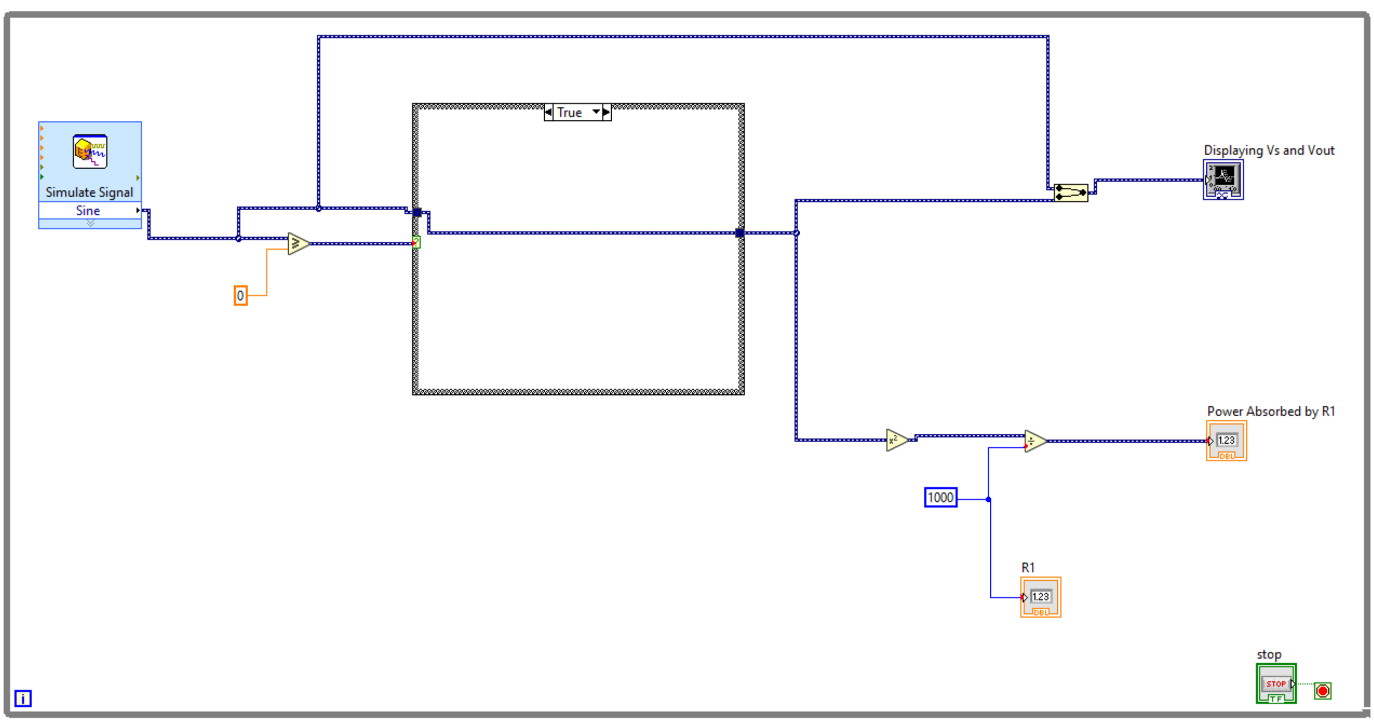
Analysis

I started by configuring a function generator (sim sig) to act as V_S. I knew it had to be sinusoidal, have an amplitude of 2, and a frequency of 0.2 Hz.

Sim Sig display:



Block Diagram:



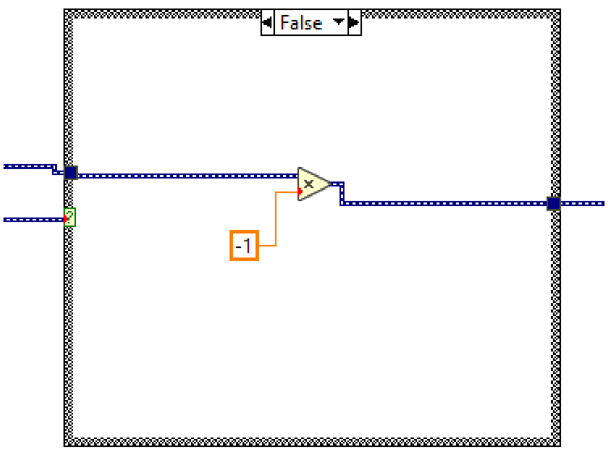
I implemented the condition of $V_S \geq 0$ by using the associated logic operator and connecting that to the "?" of the case structure to clarify the conditionals.

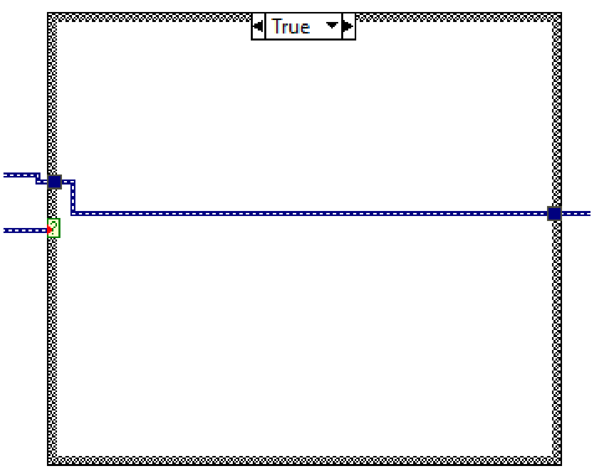
I knew I needed to display V_S and V_{out} on the same waveform chart, so I used a merge signal and connected V_S (before the case structure) and V_{out} (after the case structure).

The value of R1 was not specified, so I assumed a constant value of 1 kΩ and displayed that value to the user using a numeric indicator.

I also made a connection from V_{out} to some logic to square it and divide by R_1 (1 kΩ) to give the desired value of power absorbed by R_1. I showed this to the user using a numeric indicator.

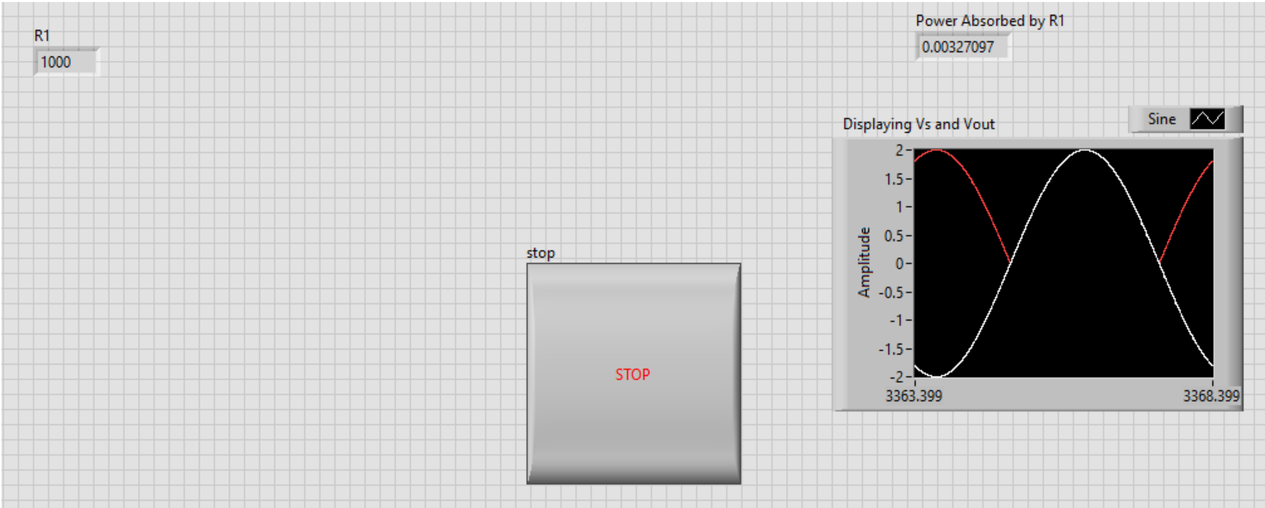
The following are pictures of the two conditions available in the case structure:





For false, it will multiply the input V_S by -1 to output $-V_S$. For true it will do nothing and just output V_S. Thus, I set the value for V_{out}.

Front Panel:



Notice how the waveforms displayed has one that has positive and negative values (V_S), and the other (V_{out}) only has positive values. This follows this conditional desired, since V_{out} multiples V_S by -1 each time its negative (ensuring positive numbers).

Conclusion

This lab was simple to implement.

Case structures are useful for operating on conditional statements and can simplify what would otherwise be rather complicated logical processes and save space on your block diagram.

Merging signals is likewise useful to see how signals look on top of each other, allowing for more direct comparisons of the two rather than guessing.

Simulated Signals are useful seeing how a real signal may act without having to use a function generator or analyzing a real signal. Sim Sig gives a surprising amount of control over all aspects of the signal.

The rest of the lab was like what we have done in previous labs.